EASYVISA
BUSINESS REPORT

Classification Model Coded Project

By

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Table of Contents

Classification Model Coded Project....1
1. Context....6
2. Objective....6
3. Data Dictionary....7
4. Data Overview through Python....7
4.1 Checking for missing values in the data....8
4.2 Checking for duplicate values in the data....9
4.3 Checking the statistical summary of the data....9
5.0 Univariate Analysis....10
5.1 Observations on no_of_employees....10
5.2 Observations on prevailing_wage....11
5.3 Observations on continent....12
5.4 Observations on education_of_employee....13
5.5 Observations on has_job_experience....14
5.6 Observations on requires_job_training....15
5.6 Observations on region_of_employment....16
5.7 Observations on unit_of_wage....17
5.8 Observations on full_time_position....18
5.9 Observations on case_status....19
6.0 Bivariate Analysis....20
6.1 Observations on case_status vs continent....21
6.2 Observations on case_status vs education_of_employee....22
6.3 Observations on case_status vs has_job_experience....23
6.4 Observations on case_status vs requires_job_training....24
6.5 Observations on case_status vs no_of_employees....25
6.5 Observations on case_status vs yr_of_estab....26
6.5 Observations on case_status vs region_of_employment....27
6.6 Observations on case_status vs prevailing_wage....28
6.7 Observations on case_status vs unit_of_wage....29
6.8 Observations on case_status vs full_time_position....30
7.0 Data Preprocessing....31

7.1 Feature Engineering 31
7.2 Checking for Outliers 31
7.3 Data Split into Train, Validation, Test sets. 32
7.4 Encoding Categorical variables. 32
8.0 Model Building 32
8.1 Model Performance – Original Data 33
8.2 Model Performance – Oversampled Data 34
8.3 Model Performance – Undersampled Data 35
8.4 Model Choosing Stage for Hyperparameter Tuning 36
9.0 Hyperparameter Tuning 36
9.1 Tuning Ada Boost Classifier model with Original data 36
9.2 Tuning Gradient Boosting classifier model with Original data 36
9.3 Tuning Gradient Boosting classifier model with Oversampled data 37
10.0 Model Comparison and Final Model Selection 37
10.1 Training data Performance Comparison 37
10.2 Validation data Performance Comparison 38
10.3 Test data Performance Comparison using selected Model 38
11.0 Feature Importance 39
12.0 Business Insights and Recommendations 40

Table of Figures

Figure 1: First 5 rows of dataset 7
Figure 2: Shape of the data 7
Figure 3: Data Types of attributes 8
Figure 4: No. of Missing values in the data 8
Figure 5: No. of Duplicate values in data 9
Figure 6: Statistical Summary of data 9
Figure 7: Box and Distribution plot 10
Figure 8: Box and Distribution plot 11
Figure 9: Count plot of continent attribute 12
Figure 10: Count plot of education_of_emplyoee 13
Figure 11: Count plot of attribute 14
Figure 12: Count plot of attribute 15
Figure 13: Count plot of attribute 16
Figure 14: Count plot of attribute 17
Figure 15: Count plot of attribute 18
Figure 16: Count plot of attribute 19
Figure 17: Heatmap of attributes 20
Figure 18: Count plot of attributes 21
Figure 19: Count plot of attributes 22
Figure 20: Count plot of attributes 23
Figure 21: Count plot of attributes 24
Figure 22: Box and Distribution plot of attributes 25
Figure 23: Box and Distribution plots of attributes 26
Figure 24: Count plot of attributes 27
Figure 25: Box and Distribution plots of attributes 28
Figure 26: Count plot of attributes 29
Figure 27: Count plot of attributes 30
Figure 28: education of employee feature engineering 31
Figure 29: Adding a derived feature. 31
Figure 30: Checking for outliers 31
Figure 31: Data split 32
Figure 32: One hot encoding of categorical variables 32
Figure 33: Training data and Validation data Performance on Original Data 33
Figure 34: Difference between Performances 33
Figure 35: Oversampled Data Shape and Description 34
Figure 36: Training data and Validation data Performance on Oversampled Data 34
Figure 37: Difference between Performances 34
Figure 38: Undersampled Data Shape and Description 35
Figure 39: Training data and Validation data Performance on Undersampled Data 35
Figure 40: Difference between Performances 35

Figure 41: Tuned Model Performance on Training data 36
Figure 42: Tuned Model Performance on Validation data 36
Figure 43: Tuned Model Performance on Training data 36
Figure 44: Tuned Model Performance on Validation data 37
Figure 45: Tuned Model Performance on Training data 37
Figure 46: Tuned Model Performance on Validation data 37
Figure 47: Model Comparison on Training Data 37
Figure 48: Model Comparison on Validation Data 38
Figure 49:Final Performance on Test Data 38
Figure 50: Feature Importance 39

1. Context

Business communities in the United States are facing high demand for human resources, but one of the constant challenges is identifying and attracting the right talent, which is perhaps the most important element in remaining competitive. Companies in the United States look for hard-working, talented, and qualified individuals both locally as well as abroad.

The Immigration and Nationality Act (INA) of the US permits foreign workers to come to the United States to work on either a temporary or permanent basis. The act also protects US workers against adverse impacts on their wages or working conditions by ensuring US employers' compliance with statutory requirements when they hire foreign workers to fill workforce shortages. The immigration programs are administered by the Office of Foreign Labor Certification (OFLC).

OFLC processes job certification applications for employers seeking to bring foreign workers into the United States and grants certifications in those cases where employers can demonstrate that there are not sufficient US workers available to perform the work at wages that meet or exceed the wage paid for the occupation in intended employment.

2. Objective

In FY 2016, the OFLC processed 775,979 employer applications for 1,699,957 positions for temporary and permanent labour certifications. This was a nine percent increase in the overall number of processed applications from the previous year. The process of reviewing every case is becoming a tedious task as the number of applicants is increasing every year.

The increasing number of applicants every year calls for a Machine Learning based solution that can help in shortlisting the candidates having higher chances of VISA approval. OFLC has hired the firm EasyVisa for data-driven solutions. You as a data scientist at EasyVisa must analyse the data provided and, with the help of a classification model:

1. Facilitate the process of visa approvals.
2. Recommend a suitable profile for the applicants for whom the visa should be certified or denied based on the drivers that significantly influence the case status.

3. Data Dictionary

The data contains the different attributes of the employee and the employer. The detailed data dictionary is given below.

- case_id: ID of each visa application
- continent: Information of continent the employee
- education_of_employee: Information of education of the employee
- has_job_experience: Does the employee has any job experience? Y= Yes; N = No
- requires_job_training: Does the employee require any job training? Y = Yes; N = No
- no_of_employees: Number of employees in the employer's company
- yr_of_estab: Year in which the employer's company was established
- region_of_employment: Information of foreign worker's intended region of employment in the US.
- prevailing_wage: Average wage paid to similarly employed workers in a specific occupation in intended employment. The purpose of the prevailing wage is to ensure that the foreign worker is not underpaid compared to other workers offering the same or similar service in the same area of employment.
- unit_of_wage: Unit of prevailing wage. Values include Hourly, Weekly, Monthly, and Yearly.
- full_time_position: Is the position of work full-time? Y = Full-Time Position; N = Part-Time Position
- case_status: Flag indicating if the Visa was certified or denied

4. Data Overview through Python

Data which is stored in csv file is imported to Jupyter Notebook with the help of pandas, NumPy libraries and used matplotlib and seaborn libraries to help with data visualization. Dataset is stored in python variable named data.

	case_id	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_v
0	EZYV01	Asia	High School	N	N	14513	2007	West	592.203	
1	EZYV02	Asia	Master's	Y	N	2412	2002	Northeast	83425.650	
2	EZYV03	Asia	Bachelor's	N	Y	44444	2008	West	122996.860	
3	EZYV04	Asia	Bachelor's	N	N	98	1897	West	83434.030	
4	EZYV05	Africa	Master's	Y	N	1082	2005	South	149907.390	

Figure 1: First 5 rows of dataset

The dataset has **25480** rows and **12** columns.

```
(25480, 12)
```

Figure 2: Shape of the data

Checking the data types of data.

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 25480 entries, 0 to 25479
Data columns (total 12 columns):
 #   Column                 Non-Null Count  Dtype
---  ------                 --------------  -----
 0   case_id                25480 non-null  object
 1   continent              25480 non-null  object
 2   education_of_employee  25480 non-null  object
 3   has_job_experience     25480 non-null  object
 4   requires_job_training  25480 non-null  object
 5   no_of_employees        25480 non-null  int64
 6   yr_of_estab            25480 non-null  int64
 7   region_of_employment   25480 non-null  object
 8   prevailing_wage        25480 non-null  float64
 9   unit_of_wage           25480 non-null  object
 10  full_time_position     25480 non-null  object
 11  case_status            25480 non-null  object
dtypes: float64(1), int64(2), object(9)
memory usage: 2.3+ MB
```

Figure 3: Data Types of attributes

4.1 Checking for missing values in the data.

```
case_id                  0
continent                0
education_of_employee    0
has_job_experience       0
requires_job_training    0
no_of_employees          0
yr_of_estab              0
region_of_employment     0
prevailing_wage          0
unit_of_wage             0
full_time_position       0
case_status              0
dtype: int64
```

- There are no null values in the dataset

Figure 4: No. of Missing values in the data

4.2 Checking for duplicate values in the data.

```
0
```

- No duplicates in the dataset.

Figure 5: No. of Duplicate values in data

4.3 Checking the statistical summary of the data.

	count	unique	top	freq	mean	std	min	25%	50%	75%	max
case_id	25480	25480	EZYV01	1	NaN	NaN	NaN	NaN	NaN	NaN	NaN
continent	25480	6	Asia	16861	NaN	NaN	NaN	NaN	NaN	NaN	NaN
education_of_employee	25480	4	Bachelor's	10234	NaN	NaN	NaN	NaN	NaN	NaN	NaN
has_job_experience	25480	2	Y	14802	NaN	NaN	NaN	NaN	NaN	NaN	NaN
requires_job_training	25480	2	N	22525	NaN	NaN	NaN	NaN	NaN	NaN	NaN
no_of_employees	25480.000	NaN	NaN	NaN	5667.043	22877.929	-26.000	1022.000	2109.000	3504.000	602069.000
yr_of_estab	25480.000	NaN	NaN	NaN	1979.410	42.367	1800.000	1976.000	1997.000	2005.000	2016.000
region_of_employment	25480	5	Northeast	7195	NaN	NaN	NaN	NaN	NaN	NaN	NaN
prevailing_wage	25480.000	NaN	NaN	NaN	74455.815	52815.942	2.137	34015.480	70308.210	107735.513	319210.270
unit_of_wage	25480	4	Year	22962	NaN	NaN	NaN	NaN	NaN	NaN	NaN
full_time_position	25480	2	Y	22773	NaN	NaN	NaN	NaN	NaN	NaN	NaN
case_status	25480	2	Certified	17018	NaN	NaN	NaN	NaN	NaN	NaN	NaN

Figure 6: Statistical Summary of data

- **Case ID**: This represents a unique identifier for each case. There are 25,480 cases, each with a unique identifier.
- **Continent**: The continent where the employment is located. There are 6 unique continents represented in this dataset. Asia has the highest frequency with 16,861 cases.
- **Education of Employee:** This indicates the highest education level attained by employees. There are 4 unique education levels recorded, with a Bachelor's degree being the most common (10,234 cases).
- **Has Job Experience:** This indicates whether employees have prior job experience. There are 2 unique responses (Yes or No). The majority, 14,802 cases, have job experience (Yes).
- **Requires Job Training:** This indicates whether the job requires any training. There are 2 unique responses (Yes or No). Most jobs do not require training, with 22,525 cases indicating No.
- **Number of Employees:** This represents the number of employees in the company. The dataset has 25,480 entries with the number of employees ranging from -26 to 602,069. The average number of employees is approximately 5,667, with a standard

deviation of 22,877. The 25th percentile is at 1,022, the median is 2,109, and the 75th percentile is at 3,504.

- **Year of Establishment**: This represents the year the company was established. There are 25,480 entries with years ranging from 1800 to 2016. The average year of establishment is around 1979, with a standard deviation of 42 years. The 25th percentile is in 1976, the median is in 1997, and the 75th percentile is in 2005.
- **Region of Employment:** This indicates the region where the employment is located. There are 5 unique regions, with the Northeast region being the most common (7,195 cases).
- **Prevailing Wage:** This represents the wage offered for the job. There are 25,480 entries with wages ranging from 2.137 to 319,210.270. The average wage is approximately 74,456, with a standard deviation of 52,816. The 25th percentile wage is 34,015.480, the median wage is 70,308.210, and the 75th percentile wage is 107,735.513.
- **Unit of Wage:** This indicates the unit of measurement for the wage (e.g., Year, Month, Hour, Week). There are 4 unique units, with the Year being the most common (22,962 cases).
- **Full-Time Position:** This indicates whether the job is a full-time position. There are 2 unique responses (Yes or No), with the majority being full-time positions (22,773 cases).
- **Case Status:** This indicates the status of the case (e.g., Certified, Denied). There are 2 unique statuses, with the majority being Certified (17,018 cases).

5.0 Univariate Analysis

5.1 Observations on no_of_employees

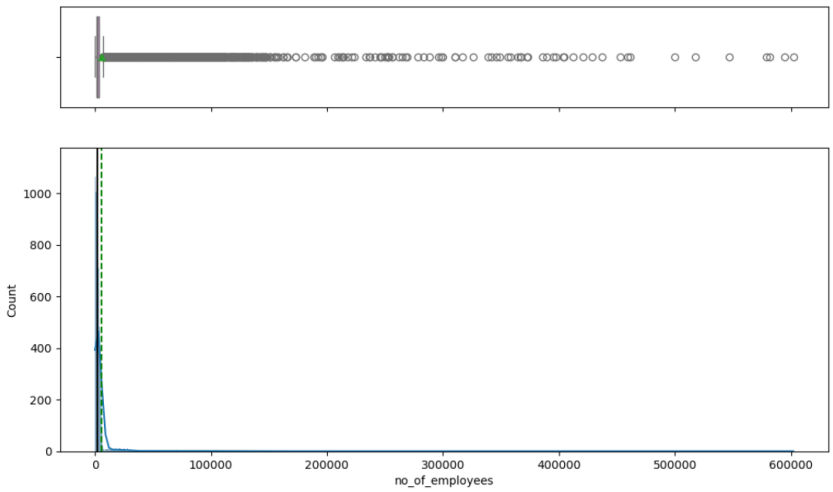

Figure 7: Box and Distribution plot

- The distribution of the number of employees is highly skewed to the right, meaning most companies have a small number of employees, but there are a few companies with a very large number of employees.
- There are many outliers in the dataset, representing these large companies.

5.2 Observations on prevailing_wage

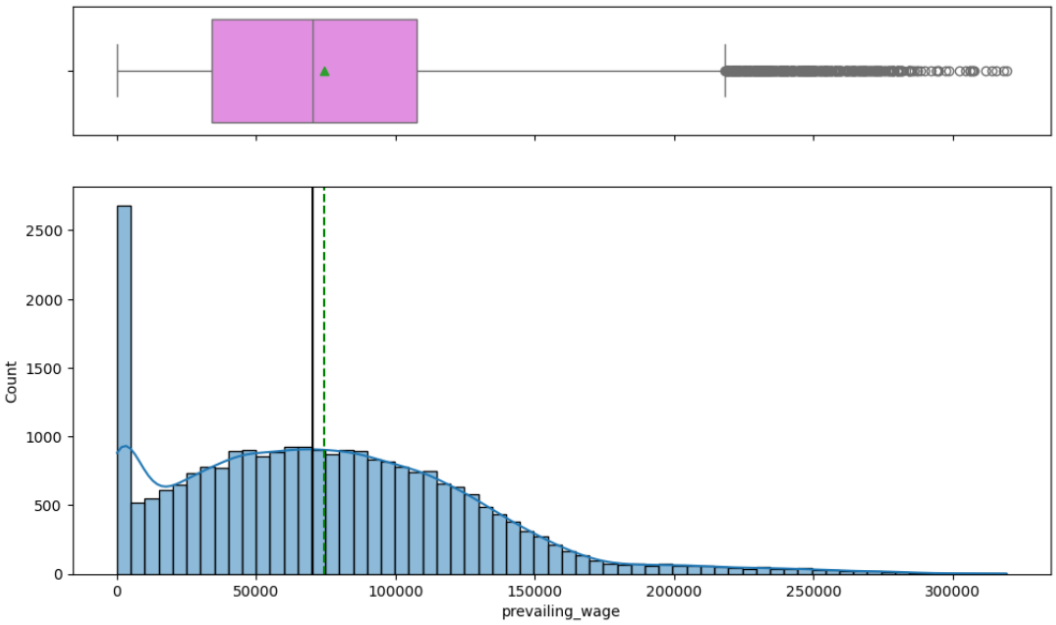

Figure 8: Box and Distribution plot

- The distribution of prevailing wages is highly right-skewed, meaning that most employees earn lower wages, while a few earn significantly higher wages.
- The dataset contains numerous outliers, suggesting that there are extreme wage values that deviate significantly from the central tendency.

5.3 Observations on continent

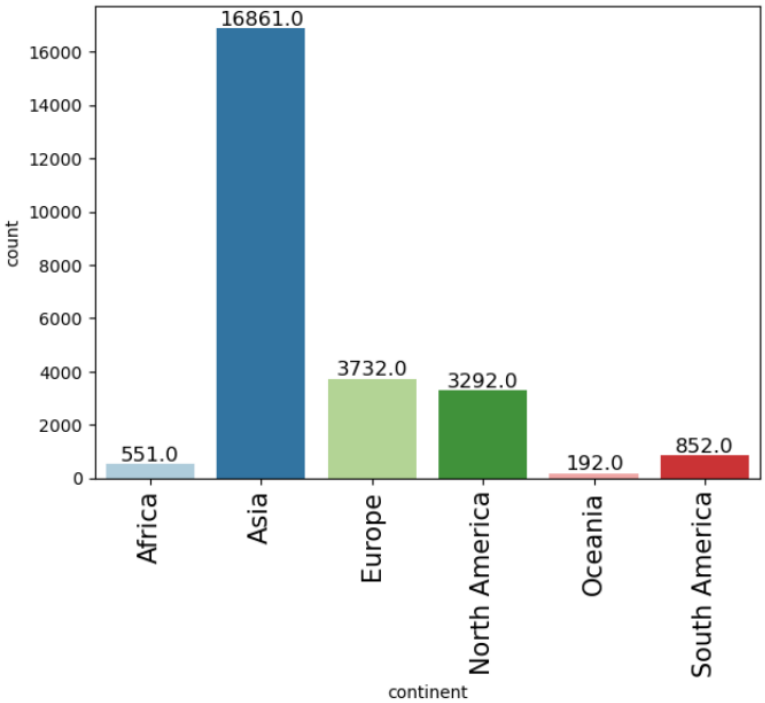

Figure 9: Count plot of continent attribute

- Asia has the highest count of data points at 16,861, indicating a significant concentration of data in this continent.
- Oceania has the lowest count at 192, suggesting that there are relatively fewer data points from this region.
- Other continents like Africa, Europe, and North America have moderate counts, with Europe and North America having similar values.

5.4 Observations on education_of_employee

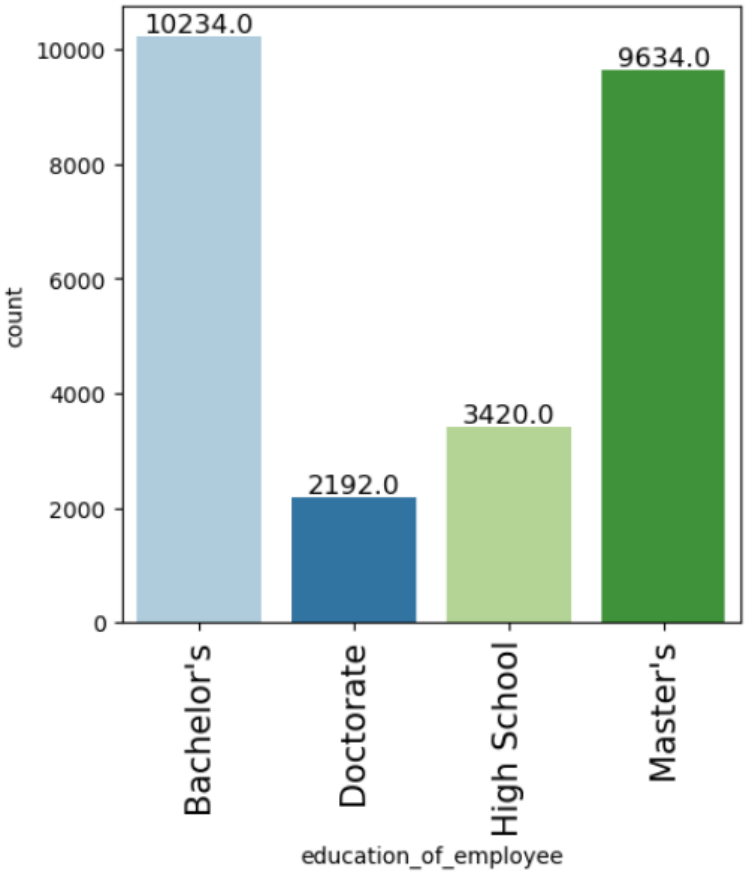

Figure 10: Count plot of education_of_emplyoee

- Most employees hold either a Bachelor's or a Master's degree.
- There are fewer employees with only a High School education, and even fewer with a Doctorate degree.

5.5 Observations on has_job_experience

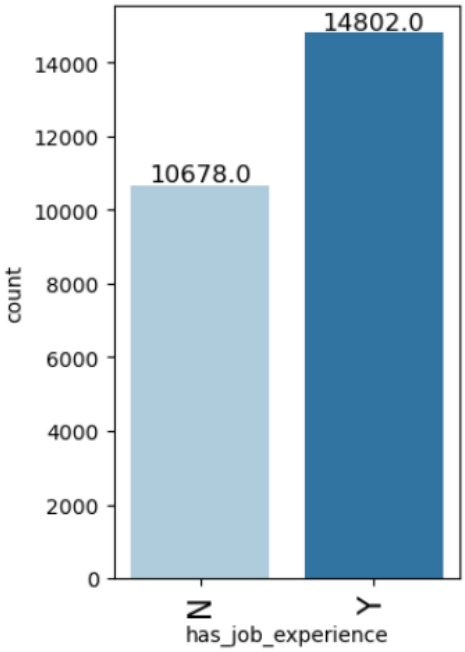

Figure 11: Count plot of attribute

- Most individuals in the dataset have job experience, with 14,802 cases falling into this category.
- A significant portion, 10,678 individuals, do not have job experience, but this number is lower compared to those with experience.

5.6 Observations on requires_job_training

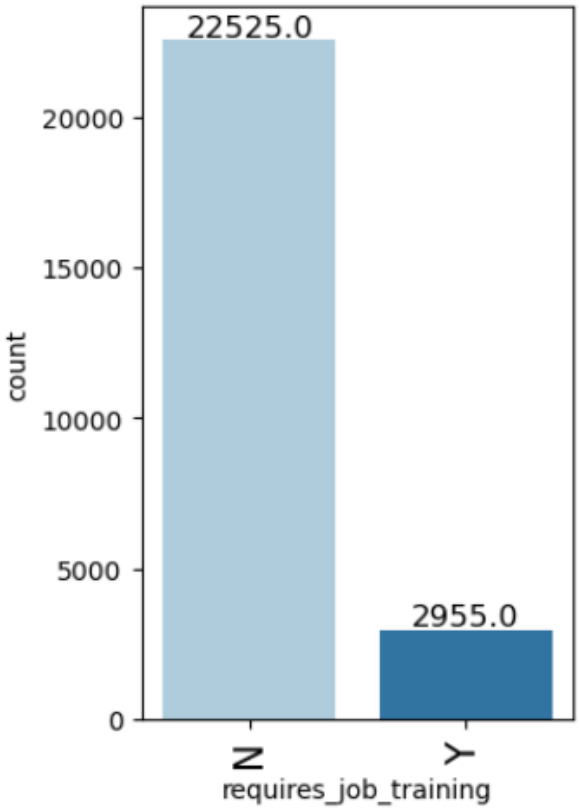

Figure 12: Count plot of attribute

- Most job positions do not require job training, as indicated by the higher count (22,525.0) in the "N" category.
- A smaller number of job positions (2,955.0) require training, highlighting that additional training is necessary for only a subset of job roles.7

5.6 Observations on region_of_employment

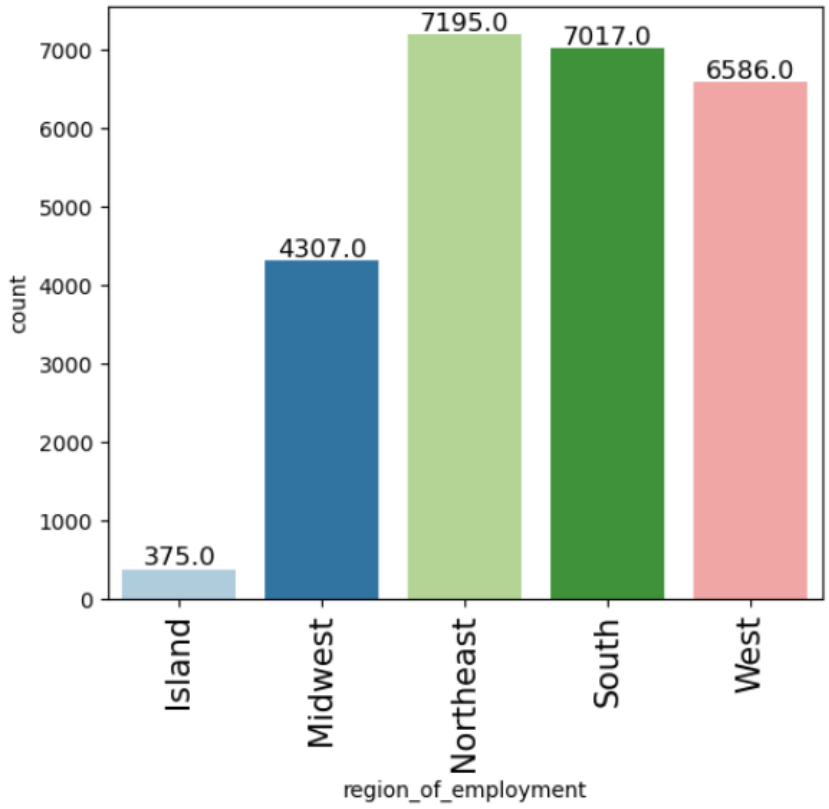

Figure 13: Count plot of attribute

- Northeast: This region has the highest count of employment at 7,195. This indicates that a significant portion of the employment data is concentrated in the Northeast region.
- South: Following closely behind, the South has a count of 7,017, also showing a strong presence of employment in this region.
- West: The West region has a substantial count of 6,586, indicating a significant employment presence.
- Midwest: The Midwest has a lower count compared to the Northeast, South, and West, with 4,307 cases.
- Island: The Island region has the lowest count of employment at 375, suggesting relatively fewer employment cases in this region.

5.7 Observations on unit_of_wage

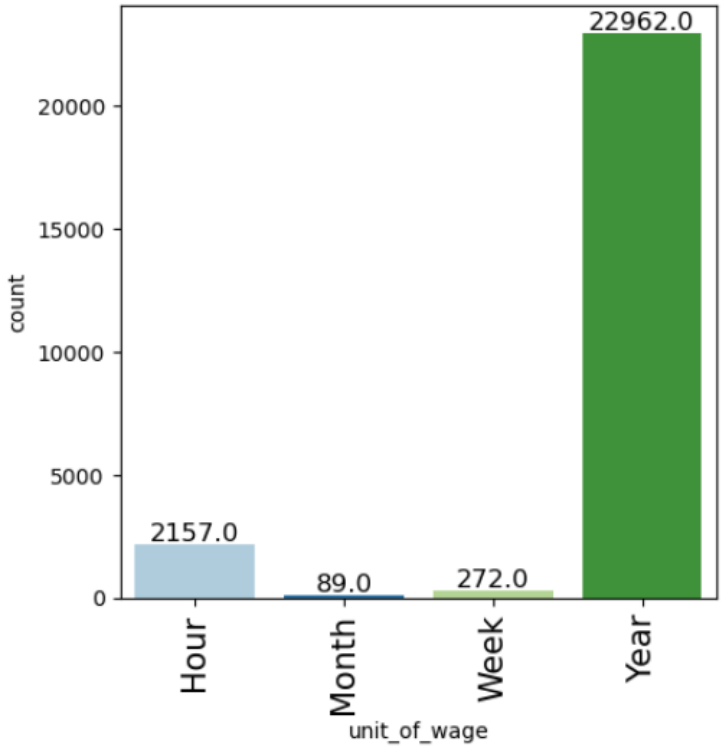

Figure 14: Count plot of attribute

- Most wages are measured on a yearly basis, with 22,962 cases. This indicates that most jobs offer annual salaries.
- The second highest count is for hourly wages, with 2,157 cases. This suggests that a notable number of jobs pay by the hour, but it is significantly less common than annual salaries.

5.8 Observations on full_time_position

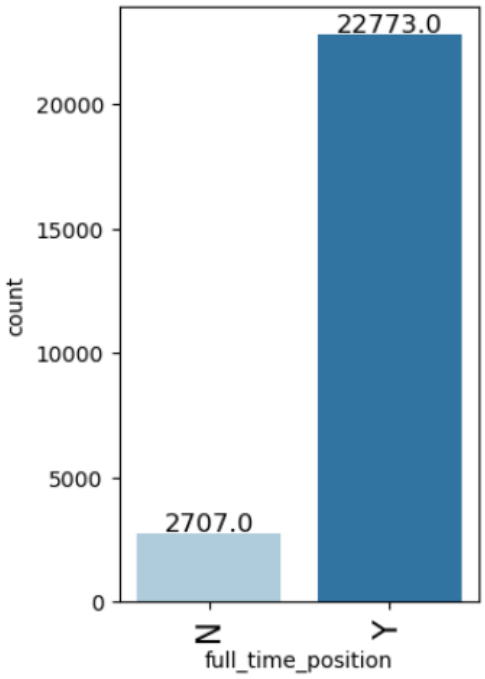

Figure 15: Count plot of attribute

- Most of the positions in the dataset are full-time, suggesting that full-time employment is the predominant work arrangement.
- The significantly lower count of non-full-time positions indicates that part-time or temporary roles are less common.

5.9 Observations on case_status

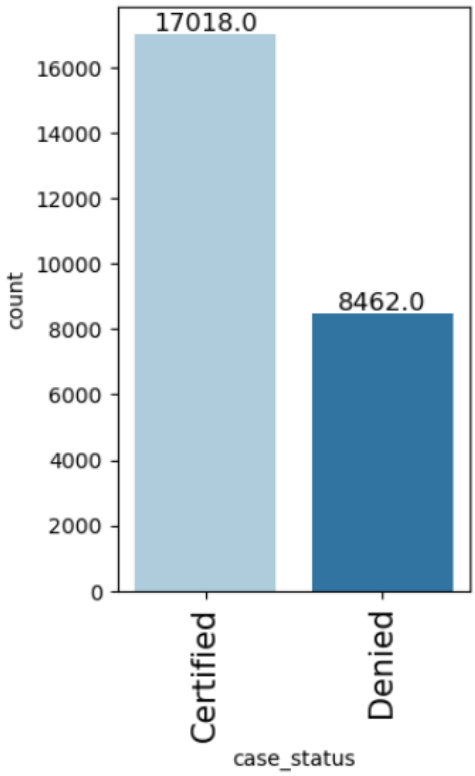

Figure 16: Count plot of attribute

- Most cases in the dataset were certified, suggesting that a higher proportion of applications met the required criteria for approval.
- There is still a significant number of denied cases, indicating that a notable portion of applications did not meet the necessary requirements.

6.0 Bivariate Analysis

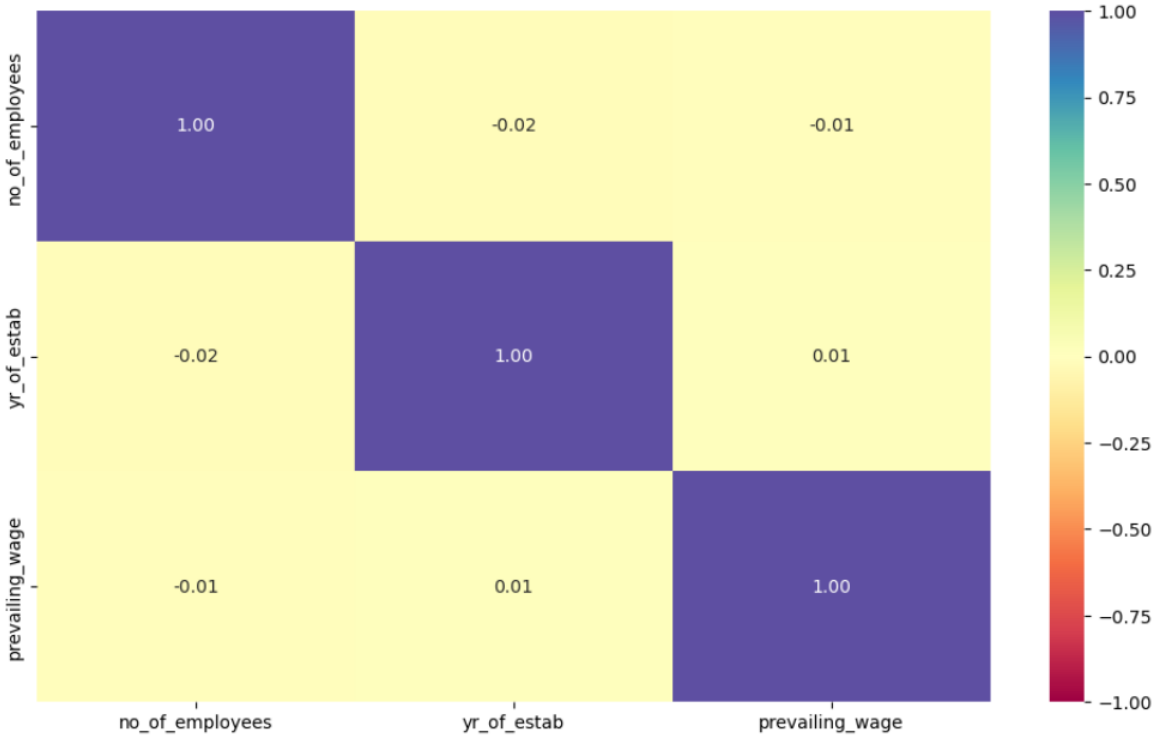

Figure 17: Heatmap of attributes

- The very weak correlations (ranging from -0.02 to 0.01) between "no_of_employees," "yr_of_estab," and "prevailing_wage" suggest that these variables do not have a significant linear relationship with each other in the dataset.
- The visualization helps in understanding that changes in one of these variables are not strongly associated with changes in the others.

6.1 Observations on case_status vs continent

case_status	Certified	Denied	All
continent			
All	17018	8462	25480
Asia	11012	5849	16861
North America	2037	1255	3292
Europe	2957	775	3732
South America	493	359	852
Africa	397	154	551
Oceania	122	70	192

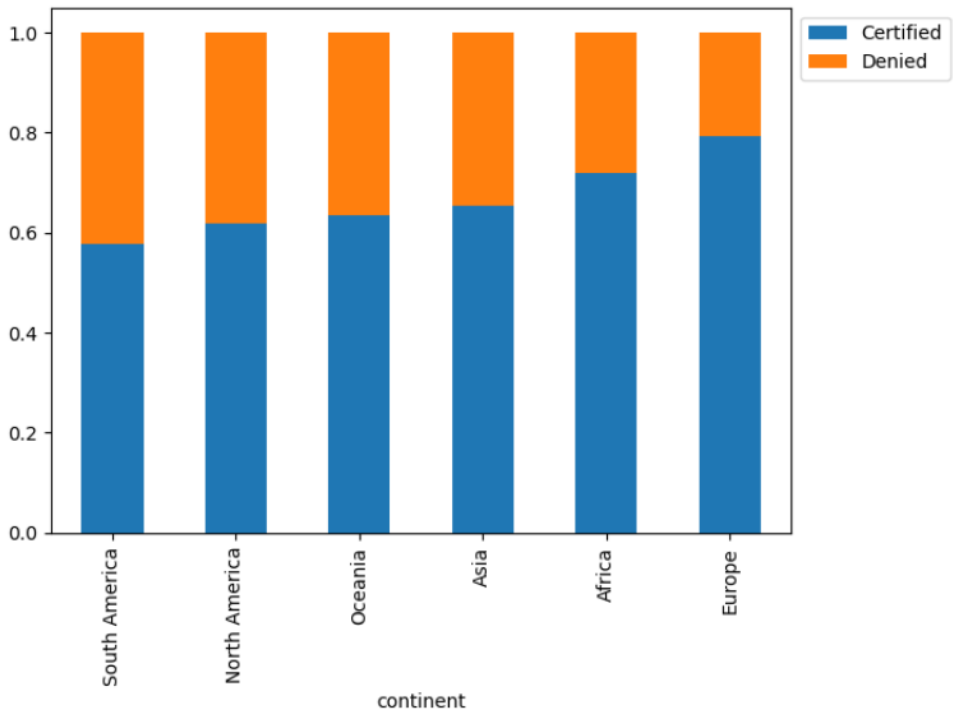

Figure 18: Count plot of attributes

- Overall Certification: Out of the total cases, the majority (17,018) were certified, indicating a higher approval rate.
- Asia: Asia has the highest number of total cases (16,861), with a majority being certified (11,012). However, it also has a significant number of denied cases (5,849).
- North America and Europe: Both continents have relatively high certification rates, with Europe having a higher proportion of certified cases compared to North America.
- South America, Africa, and Oceania: These continents have lower total case counts, with a majority being certified. South America and Africa have a notable number of denied cases, while Oceania has the least number of both certified and denied cases.

6.2 Observations on case_status vs education_of_employee

```
case_status            Certified  Denied    All
education_of_employee
All                        17018    8462  25480
Bachelor's                  6367    3867  10234
High School                 1164    2256   3420
Master's                    7575    2059   9634
Doctorate                   1912     280   2192
-----------------------------------------------------------------------
```

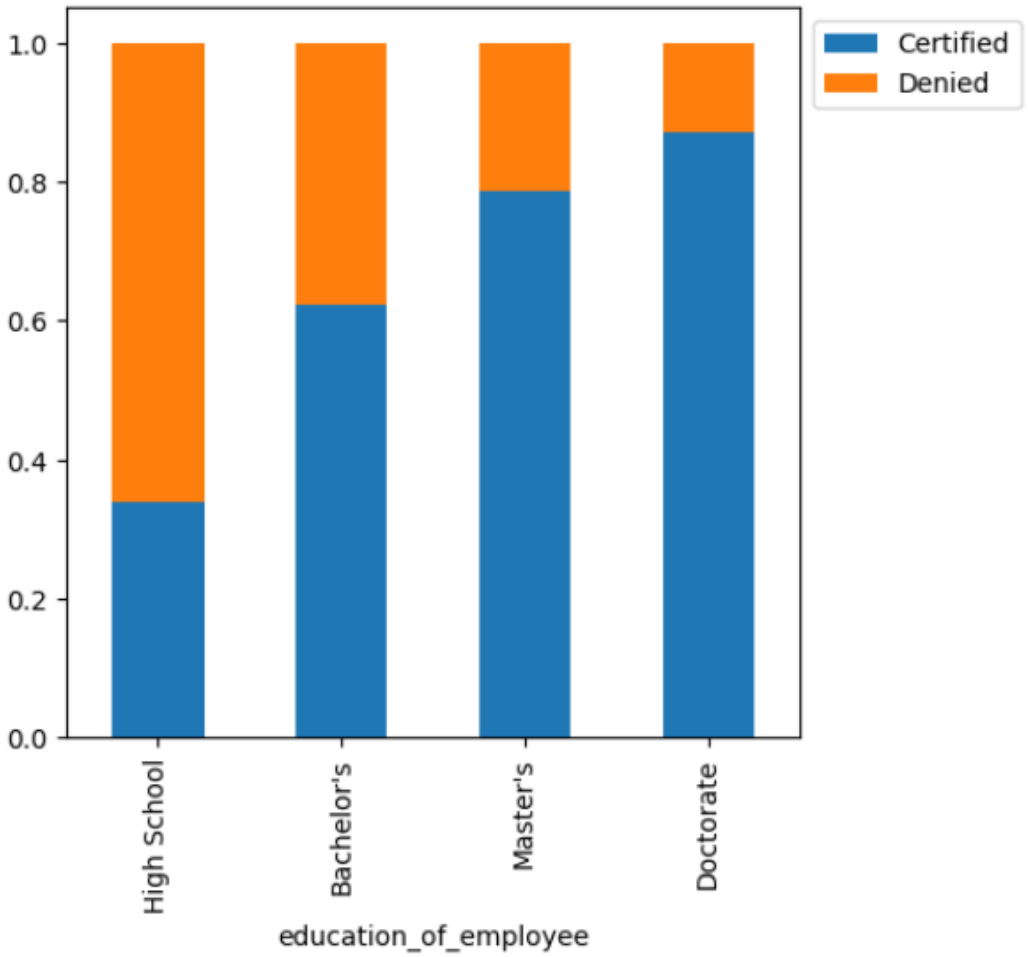

Figure 19: Count plot of attributes

- Certification Rate: The certification rate increases with higher education levels. High School has the highest denial rate, while Doctorate has the lowest denial rate.
- Bachelor's and Master's: The counts for Bachelor's and Master's degrees are relatively high, with a significant proportion of certified cases.
- High School: The denial rate is quite high for employees with only a High School education, indicating that higher education levels are more likely to get certified.
- Doctorate: Employees with a Doctorate degree have the highest certification rate and the lowest number of denials, suggesting that higher qualifications have a positive impact on certification.

6.3 Observations on case_status vs has_job_experience

```
case_status         Certified  Denied    All
has_job_experience
All                     17018    8462  25480
N                        5994    4684  10678
Y                       11024    3778  14802
-------------------------------------------------------
```

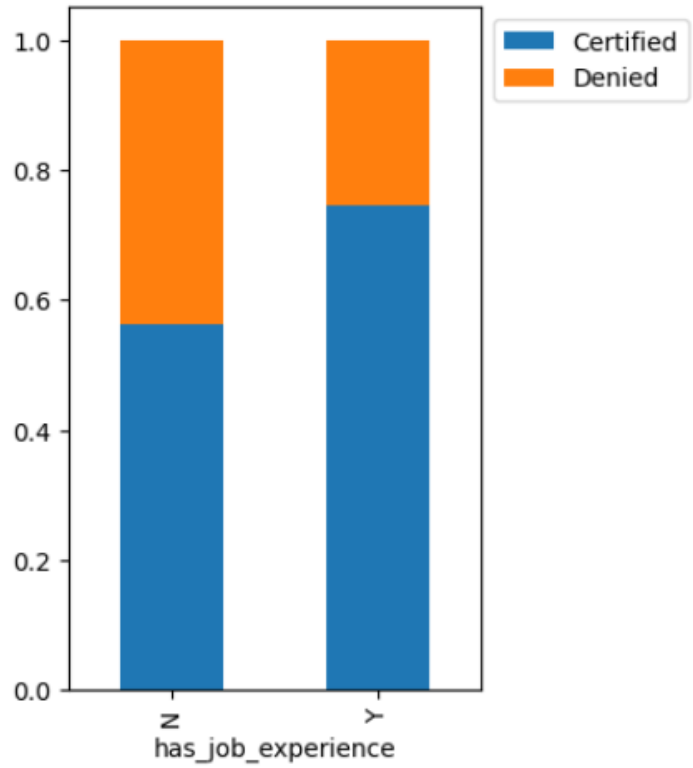

Figure 20: Count plot of attributes

- Certification Rate by Job Experience:
  - Applicants with job experience (Y) have a higher certification rate (11,024) compared to those without job experience (N), who have a lower certification rate (5,994).
  - This indicates that having job experience increases the likelihood of getting certified.
- Denial Rate by Job Experience:
  - Applicants without job experience (N) have a higher denial rate (4,684) compared to those with job experience (Y), who have a lower denial rate (3,778).
  - This suggests that lack of job experience increases the chances of application denial.
- Overall Counts:
  - The total number of cases is higher for applicants with job experience (14,802) compared to those without (10,678).

6.4 Observations on case_status vs requires_job_training

case_status	Certified	Denied	All
requires_job_training			
All	17018	8462	25480
N	15012	7513	22525
Y	2006	949	2955

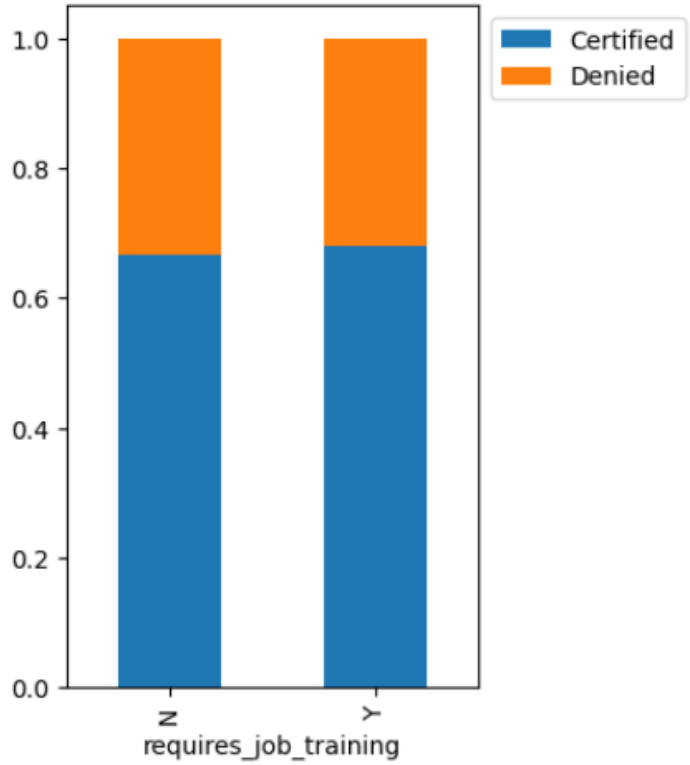

Figure 21: Count plot of attributes

- Certification Rate:
  - Jobs that do not require training (N) have a higher certification rate compared to those that do require training (Y). Out of 22,525 cases where no job training is required, 15,012 (approximately 67%) were certified.
  - In contrast, out of 2,955 cases where job training is required, 2,006 (approximately 68%) were certified.
- Denial Rate:
  - Jobs requiring no training (N) had 7,513 cases denied.
  - Jobs requiring training (Y) had 949 cases denied.
  - This suggests that most denied cases are for jobs that do not require training.
- Overall Distribution:
  - Most job cases do not require training (22,525), compared to the smaller number of cases that require training (2,955).

- Both categories (N and Y) have a higher proportion of certifications than denials, indicating an overall tendency towards case approval.

6.5 Observations on case_status vs no_of_employees

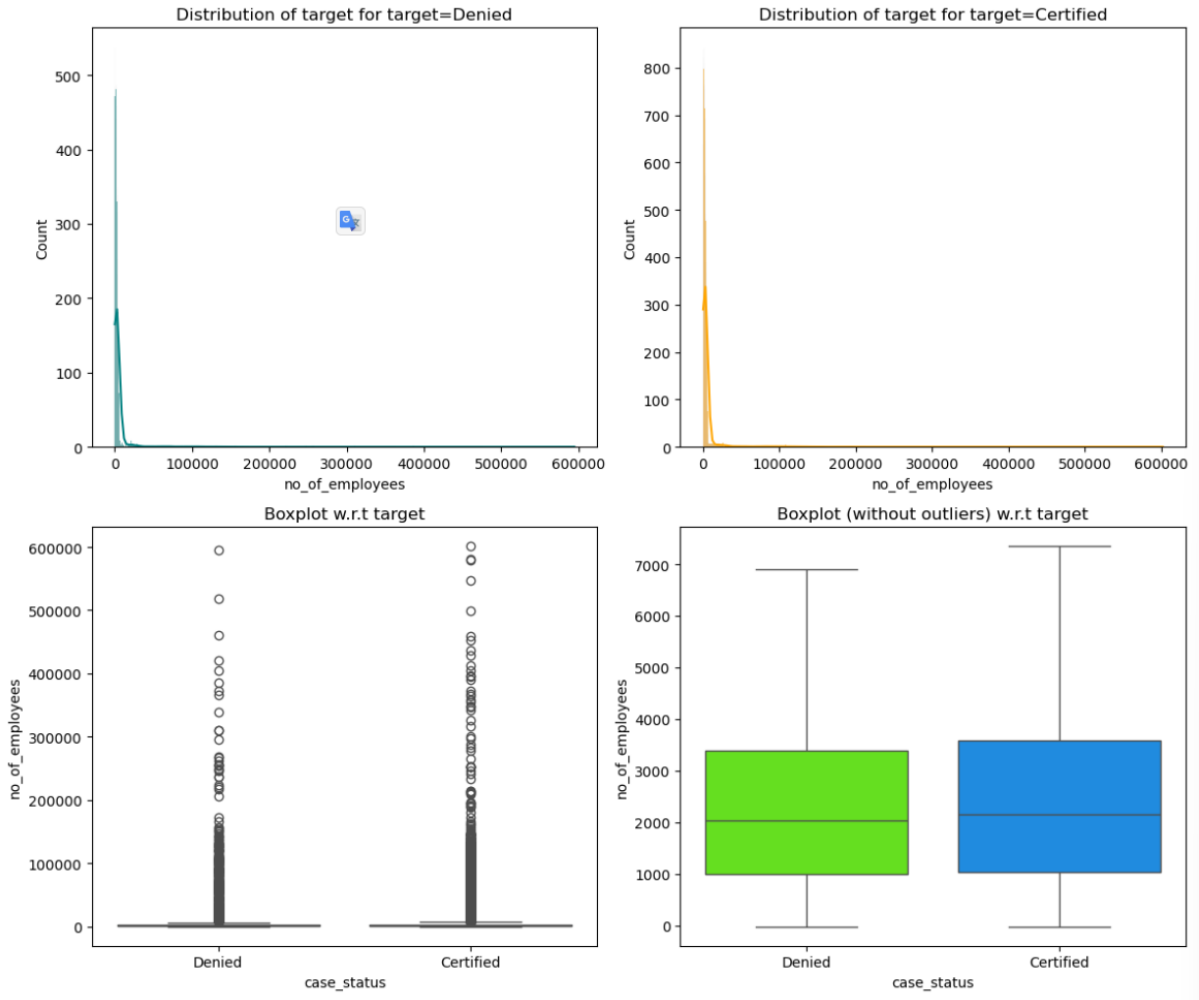

Figure 22: Box and Distribution plot of attributes

- Most of the cases, whether denied or certified, involve companies with a smaller number of employees.
- There are a few larger companies with a high number of employees, but they are outliers in the dataset.
- The central tendency (median) is similar for both denied and certified cases, suggesting that company size alone may not be a significant differentiator for case status.

6.5 Observations on case_status vs yr_of_estab

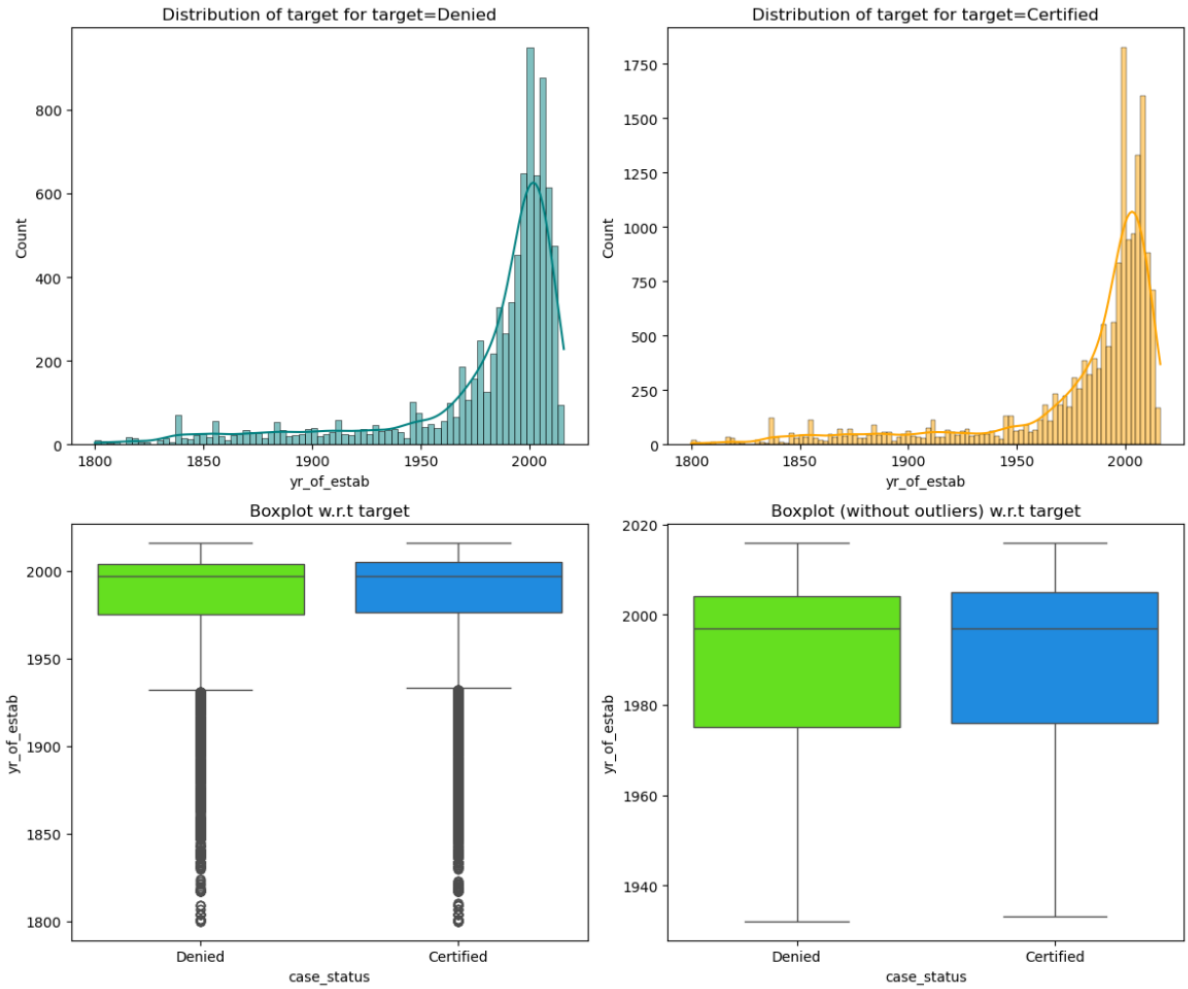

Figure 23: Box and Distribution plots of attributes

- Both denied and certified cases show an increasing trend over time, with a peak around the year 2000.
- The median year of establishment for both denied and certified cases is similar, around the year 2000.
- The interquartile range is also similar for both denied and certified cases, indicating that the year of establishment is not a significant differentiator for case status.
- The presence of outliers in the bottom-left boxplot suggests there are some cases with an unusual year of establishment, but the overall trend remains consistent.

6.5 Observations on case_status vs region_of_employment

case_status	Certified	Denied	All
region_of_employment			
All	17018	8462	25480
Northeast	4526	2669	7195
West	4100	2486	6586
South	4913	2104	7017
Midwest	3253	1054	4307
Island	226	149	375

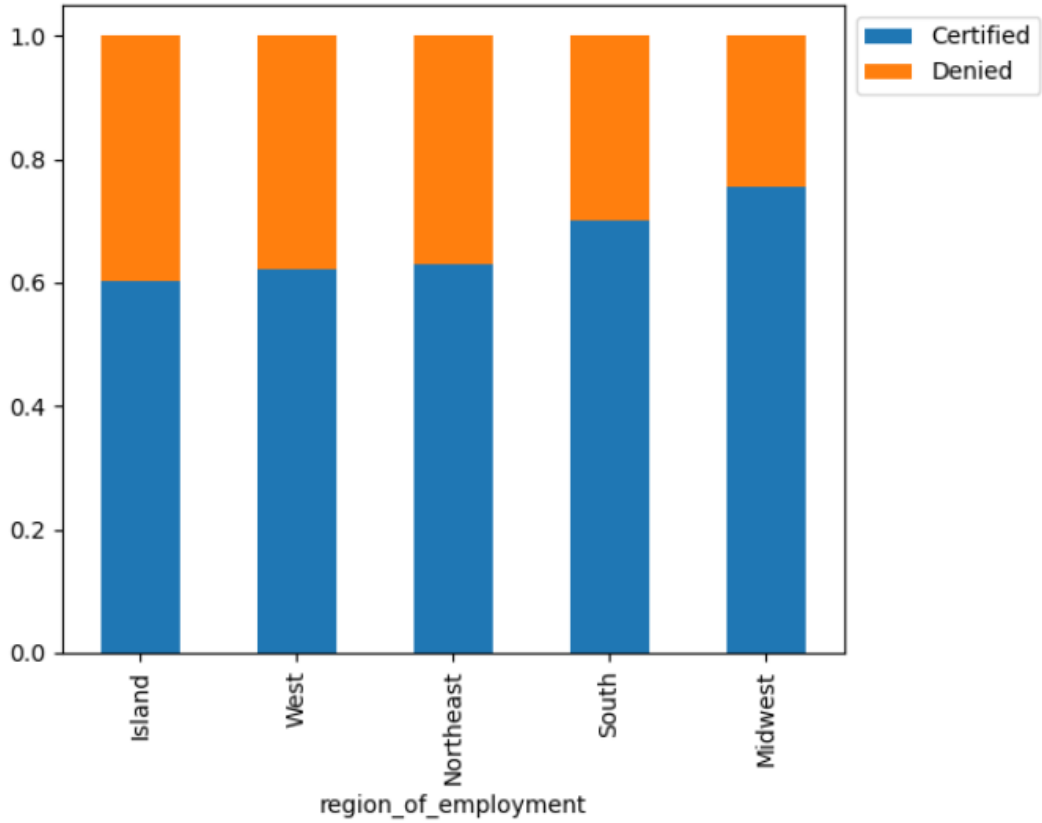

Figure 24: Count plot of attributes

- Northeast: This region has the highest count of employment cases (7,195), with a significant number of certified cases (4,526) and denied cases (2,669).
- West: The West region has a considerable number of employment cases (6,586), with more certified cases (4,100) than denied cases (2,486).
- South: The South region also has a high count of employment cases (7,017), with a majority being certified (4,913) and a smaller number being denied (2,104).
- Midwest: The Midwest has a moderate count of employment cases (4,307), with more certified cases (3,253) compared to denied cases (1,054).
- Island: The Island region has the lowest count of employment cases (375), with a majority being certified (226) and a smaller number being denied (149).

6.6 Observations on case_status vs prevailing_wage

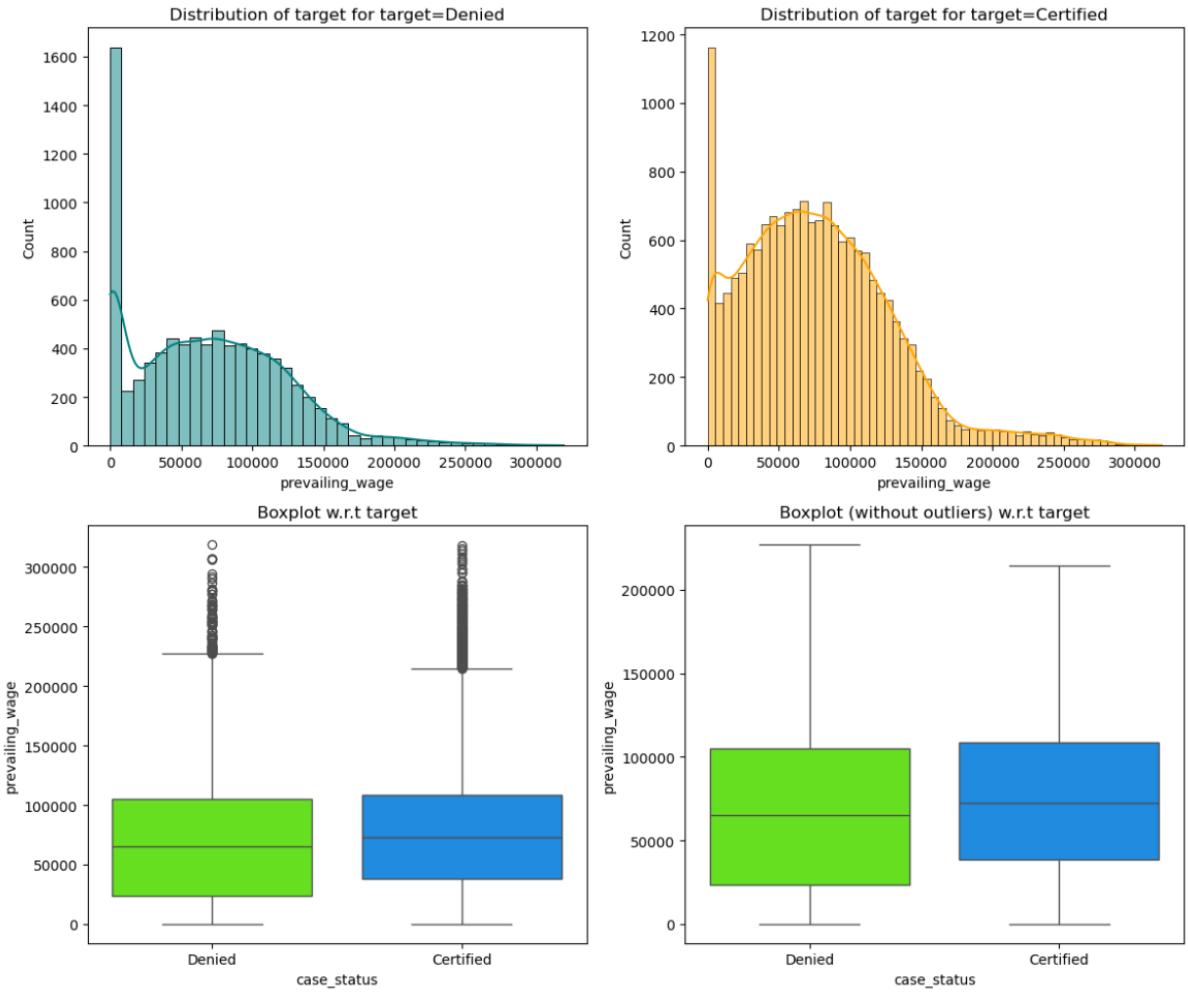

Figure 25: Box and Distribution plots of attributes

- Both the denied and certified cases show a right-skewed distribution of prevailing wages, with most cases concentrated at lower wage values.
- The boxplots indicate that the median prevailing wage is similar for both denied and certified cases.
- The presence of outliers in the bottom left plot suggests there are extreme wage values in the dataset.

6.7 Observations on case_status vs unit_of_wage

```
case_status   Certified  Denied    All
unit_of_wage
All               17018    8462  25480
Year              16047    6915  22962
Hour                747    1410   2157
Week                169     103    272
Month                55      34     89
--------------------------------------------------------------------------
```

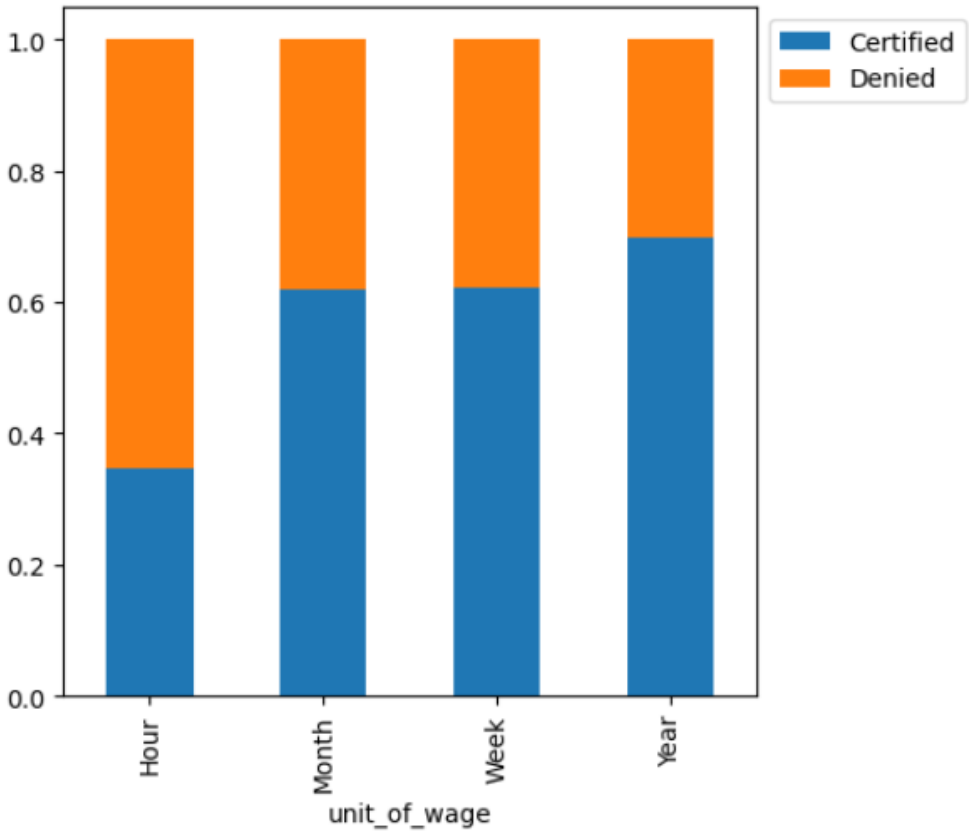

Figure 26: Count plot of attributes

- Year: Many cases (22,962) use "Year" as the unit of wage. Among these, 16,047 are certified, and 6,915 are denied. This indicates that annual wages are the most common and have a high certification rate.
- Hour: There are 2,157 cases using "Hour" as the unit of wage, with 747 certified and 1,410 denied. This suggests that hourly wages are less common and have a higher denial rate.
- Week: There are 272 cases using "Week" as the unit of wage, with 169 certified and 103 denied. Weekly wages are relatively rare but have a decent certification rate.
- Month: There are 89 cases using "Month" as the unit of wage, with 55 certified and 34 denied. Monthly wages are the least common and have a moderate certification rate.

6.8 Observations on case_status vs full_time_position

case_status	Certified	Denied	All
full_time_position			
All	17018	8462	25480
Y	15163	7610	22773
N	1855	852	2707

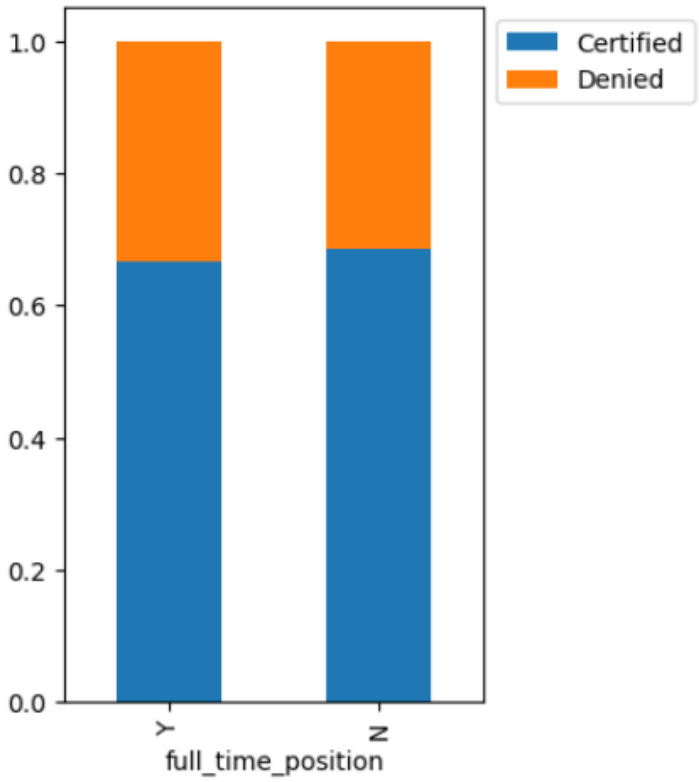

Figure 27: Count plot of attributes

- Full-Time Positions (Y): Many cases (22,773) are for full-time positions. Among these, 15,163 are certified, and 7,610 are denied. This indicates that full-time positions have a high certification rate.
- Non-Full-Time Positions (N): There are 2,707 cases for non-full-time positions, with 1,855 certified and 852 denied. Although the numbers are smaller, the certification rate is still relatively high.

7.0 Data Preprocessing

7.1 Feature Engineering

7.1.1 Dealing with Ordinal data point education_of_employee

	continent	education_of_employee	has_job_experience	requires_job_training	no_of_employees	yr_of_estab	region_of_employment	prevailing_wage	unit_of_wage
0	Asia	0	N	N	14513	2007	West	592.203	Hour
1	Asia	2	Y	N	2412	2002	Northeast	83425.650	Year
2	Asia	1	N	Y	44444	2008	West	122996.860	Year
3	Asia	1	N	N	98	1897	West	83434.030	Year
4	Africa	2	Y	N	1082	2005	South	149907.390	Year

Figure 28: education of employee feature engineering

7.1.2 adding a derived feature named Company Age from yr_of_estab

loyee	has_job_experience	requires_job_training	no_of_employees	region_of_employment	prevailing_wage	unit_of_wage	full_time_position	case_status	company_age
0	N	N	14513	West	592.203	Hour	Y	Denied	18
2	Y	N	2412	Northeast	83425.650	Year	Y	Certified	23
1	N	Y	44444	West	122996.860	Year	Y	Denied	17
1	N	N	98	West	83434.030	Year	Y	Denied	128
2	Y	N	1082	South	149907.390	Year	Y	Certified	20

Figure 29: Adding a derived feature.

7.2 Checking for Outliers

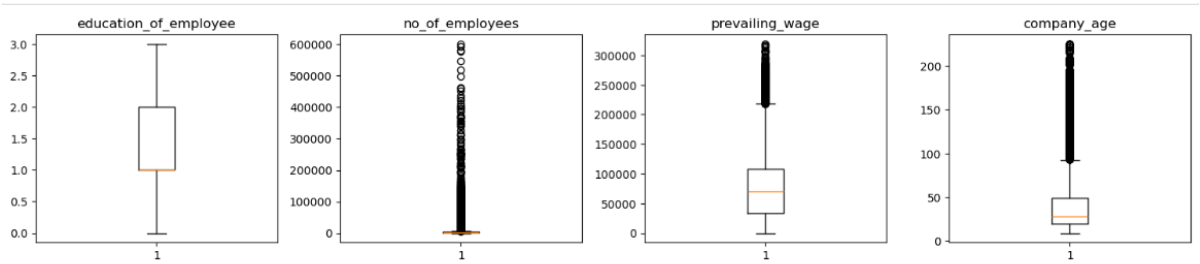

Figure 30: Checking for outliers

Each of the 4 numerical attributes contain outliers but these are real values which are true in nature in Realtime. So, treating them would be manipulating the data which in turn effects the model performance.

7.3 Data Split into Train, Validation, Test sets.

```
Number of rows in train data = 15288
Number of rows in validation data = 5096
Number of rows in test data = 5096
```

Figure 31: Data split

7.4 Encoding Categorical variables.

```
(15288, 19) (5096, 19) (5096, 19)
```

- After encoding there are 19 columns.

Figure 32: One hot encoding of categorical variables

8.0 Model Building

Model evaluation criterion

Model can make wrong predictions as:

Predicting an applicant will get visa approved, the good applicant is Certified

Predicting the bad applicant is Certified will have negative impact on model.

Which case is more important?

Predicting an applicant whose profile is bad but the model predicted as certified will impact the visa process.

How to reduce this loss i.e., need to reduce False Negatives?

OFLC would want Recall to be maximized, greater the Recall lesser the chances of false negatives.

8.1 Model Performance – Original Data

```
Training Performance:

Bagging: 0.9868756121449559
Random forest: 1.0
GBM: 0.8790401567091087
Adaboost: 0.890205680705191
dtree: 1.0

Validation Performance:

Bagging: 0.781433607520564
Random forest: 0.8419506462984724
GBM: 0.8769095182138661
Adaboost: 0.882491186839013
dtree: 0.7564629847238543
```

Figure 33: Training data and Validation data Performance on Original Data

8.1.1 Training and Validation Performance Difference: Original Data

```
Training and Validation Performance Difference:

Bagging: Training Score: 0.9869, Validation Score: 0.7814, Difference: 0.2054
Random forest: Training Score: 1.0000, Validation Score: 0.8420, Difference: 0.1580
GBM: Training Score: 0.8790, Validation Score: 0.8769, Difference: 0.0021
Adaboost: Training Score: 0.8902, Validation Score: 0.8825, Difference: 0.0077
dtree: Training Score: 1.0000, Validation Score: 0.7565, Difference: 0.2435
```

- AdaBoost has the best performance followed by GBM model as per the validation performance

Figure 34: Difference between Performances

8.2 Model Performance – Oversampled Data

8.2.1 Oversampled Data Shape and Description

```
Before Oversampling, counts of label 'Yes': 10210
Before Oversampling, counts of label 'No': 5078

After Oversampling, counts of label 'Yes': 10210
After Oversampling, counts of label 'No': 10210

After Oversampling, the shape of train_X: (20420, 19)
After Oversampling, the shape of train_y: (20420,)
```

Figure 35: Oversampled Data Shape and Description

```
Training Performance:

Bagging: 0.9821743388834476
Random forest: 0.9999020568070519
GBM: 0.8687561214495593
Adaboost: 0.808619000979432
dtree: 1.0

Validation Performance:

Bagging: 0.7488249118683902
Random forest: 0.8064042303172738
GBM: 0.86310223266745
Adaboost: 0.8037602820211516
dtree: 0.7300235017626322
```

Figure 36: Training data and Validation data Performance on Oversampled Data

8.2.1 Training and Validation Performance Difference: Oversampled Data

```
Training and Validation Performance Difference:

Bagging: Training Score: 0.9822, Validation Score: 0.7488, Difference: 0.2333
Random forest: Training Score: 0.9999, Validation Score: 0.8064, Difference: 0.1935
GBM: Training Score: 0.8688, Validation Score: 0.8631, Difference: 0.0057
Adaboost: Training Score: 0.8086, Validation Score: 0.8038, Difference: 0.0049
dtree: Training Score: 1.0000, Validation Score: 0.7300, Difference: 0.2700
```

- GBM has the best performance on validation followed by Adaboost

Figure 37: Difference between Performances

8.3 Model Performance – Undersampled Data

8.3.1 Undersampled Data Shape and Description

```
Before Under Sampling, counts of label 'Yes': 10210
Before Under Sampling, counts of label 'No': 5078

After Under Sampling, counts of label 'Yes': 5078
After Under Sampling, counts of label 'No': 5078

After Under Sampling, the shape of train_X: (10156, 19)
After Under Sampling, the shape of train_y: (10156,)
```

Figure 38: Undersampled Data Shape and Description

```
Training Performance:

Bagging: 0.9680976762504924
Random forest: 1.0
GBM: 0.7504923198109492
Adaboost: 0.7002756990941316
dtree: 1.0

Validation Performance:

Bagging: 0.6139835487661575
Random forest: 0.6753819036427732
GBM: 0.7314923619271445
Adaboost: 0.6982961222091657
dtree: 0.6304347826086957
```

Figure 39: Training data and Validation data Performance on Undersampled Data

8.3.1 Training and Validation Performance Difference: Undersampled Data

```
Training and Validation Performance Difference:

Bagging: Training Score: 0.9681, Validation Score: 0.6140, Difference: 0.3541
Random forest: Training Score: 1.0000, Validation Score: 0.6754, Difference: 0.3246
GBM: Training Score: 0.7505, Validation Score: 0.7315, Difference: 0.0190
Adaboost: Training Score: 0.7003, Validation Score: 0.6983, Difference: 0.0020
dtree: Training Score: 1.0000, Validation Score: 0.6304, Difference: 0.3696
```

- GBM has the best performance followed by AdaBoost model as per the validation performance

Figure 40: Difference between Performances

8.4 Model Choosing Stage for Hyperparameter Tuning

After building 15 models, it was observed that both the GBM and Ada boost models, trained on an Original dataset, as well as the GBM model trained on an oversampled dataset, exhibited strong performance on both the training and validation datasets.

9.0 Hyperparameter Tuning

9.1 Tuning Ada Boost Classifier model with Original data

9.1.1 Tuned Model Performance on Train and Validation Dataset

	Accuracy	Recall	Precision	F1
0	0.733	0.908	0.747	0.819

Figure 41: Tuned Model Performance on Training data

	Accuracy	Recall	Precision	F1
0	0.737	0.908	0.750	0.822

Figure 42: Tuned Model Performance on Validation data

After Tuning Recall Score has increased from 89% to 90%.

9.2 Tuning Gradient Boosting classifier model with Original data

9.2.1 Tuned Model Performance on Train and Validation Dataset

	Accuracy	Recall	Precision	F1
0	0.737	0.914	0.748	0.823

Figure 43: Tuned Model Performance on Training data

	Accuracy	Recall	Precision	F1
0	0.741	0.912	0.752	0.825

Figure 44: Tuned Model Performance on Validation data

After Tuning Recall Score has increased from 87% to 91%.

9.3 Tuning Gradient Boosting classifier model with Oversampled data

9.3.1 Tuned Model Performance on Train and Validation Dataset

	Accuracy	Recall	Precision	F1
0	0.797	0.854	0.766	0.808

Figure 45: Tuned Model Performance on Training data

	Accuracy	Recall	Precision	F1
0	0.725	0.829	0.775	0.801

Figure 46: Tuned Model Performance on Validation data

After Tuning there is no much change in Recall Score.

10.0 Model Comparison and Final Model Selection

10.1 Training data Performance Comparison

```
Training performance comparison:
```

	AdaBoost boosting trained with Original data	Gradient boosting trained with Original data	Gradient trained with Oversampled data
Accuracy	0.733	0.737	0.797
Recall	0.908	0.914	0.854
Precision	0.747	0.748	0.766
F1	0.819	0.823	0.808

Figure 47: Model Comparison on Training Data

the Gradient Boosting model trained with original data achieves the highest recall and F1 score, indicating a strong performance in correctly identifying true positives and maintaining a balance between precision and recall.

10.2 Validation data Performance Comparison

Validation performance comparison:

	AdaBoost boosting trained with Original data	Gradient boosting trained with Original data	Gradient boosting trained with Oversampled data
Accuracy	0.737	0.741	0.725
Recall	0.908	0.912	0.829
Precision	0.750	0.752	0.775
F1	0.822	0.825	0.801

Figure 48: Model Comparison on Validation Data

The Gradient Boosting model trained with original data generally performs the best across most metrics (Accuracy, Recall, and F1 score), making it a strong candidate for further evaluation.

10.3 Test data Performance Comparison using selected Model

	Accuracy	Recall	Precision	F1
0	0.728	0.911	0.741	0.817

Figure 49:Final Performance on Test Data

- The Gradient boost model trained on Original data has given ~91% recall on the test set
- This performance is in line with what we achieved with this model on the train and validation sets
- So, this is a generalized model

11.0 Feature Importance

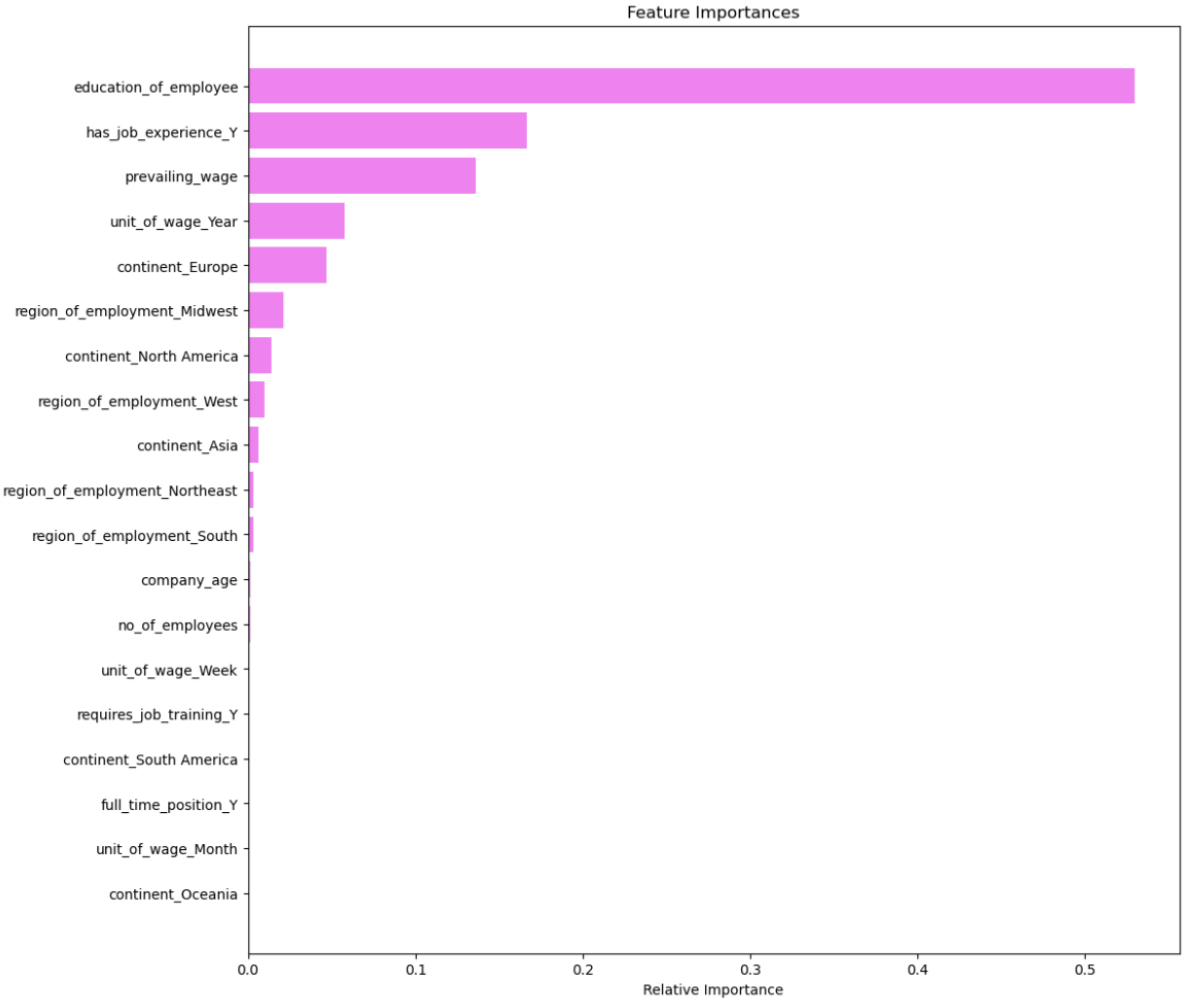

Figure 50: Feature Importance

- Top Features: The education level of employees, job experience, and prevailing wage are the most important features in the model. These factors significantly influence the predictions.
- Geographical Factors: Continent and region of employment also play a role in the model's decisions, with Europe and the Midwest having higher importance compared to other regions.
- Wage Units: The unit of wage (Year, Week, Month) has varying importance, with annual wages being more significant.
- Other Factors: Company age, number of employees, and job training requirements have lower importance but still contribute to the model's predictions.

12.0 Business Insights and Recommendations

1. Education of Employee:

Insight: Higher education levels significantly increase the likelihood of visa approval.

Recommendation: Encourage applicants to highlight their highest educational qualifications and consider partnering with educational institutions to attract highly qualified candidates.

2. Job Experience:

Insight: Applicants with prior job experience have a higher chance of visa approval.

Recommendation: Emphasize the importance of relevant job experience in the application process. Provide resources or training to help applicants gain relevant experience before applying.

3. Prevailing Wage:

Insight: Higher prevailing wages correlate with higher approval rates.

Recommendation: Ensure that wages offered are competitive and meet or exceed industry standards. Highlight the offered wage in the application to strengthen the case.

4. Unit of Wage:

Insight: Yearly wages are preferred.

Recommendation: Whenever possible, offer wages on a yearly basis to increase the likelihood of visa approval. Clearly specify the yearly wage in the application.

5. Continent of Origin:

Insight: Applicants from specific continents, such as Europe, have different approval rates.

Recommendation: Tailor recruitment strategies based on the continent of origin. For example, target regions with higher approval rates for sourcing talent.

6. Region of Employment:

Insight: The intended region of employment within the US affects approval chances.

Recommendation: Focus on regions with higher demand for foreign workers and higher approval rates. Provide guidance to applicants on selecting regions with better approval prospects.

7. Full-Time Position:

Insight: Full-time positions have higher approval rates compared to part-time positions.

Recommendation: Whenever possible, offer full-time positions to increase the likelihood of visa approval. Clearly specify the full-time nature of the position in the application.

8. Company Age:

Insight: Older, more established companies have higher approval rates.

Recommendation: Highlight the stability and reputation of the employer's company in the application. For newer companies, provide additional documentation to demonstrate stability and growth potential.

9. Number of Employees:

Insight: Larger companies tend to have higher approval rates.

Recommendation: Highlight the size and resources of the employer's company to strengthen the application. Smaller companies can focus on showcasing their growth trajectory and resource management.